

RUDRO CHAKRABORTY

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Professional Summary

Dedicated Computer Science student (B.Tech) specializing in full-stack web development, API infrastructure, embedded systems, and AI integration. Experienced in translating foundational programming concepts into functional, real-world projects, ranging from custom secure database engines to analytical enterprise software. A proactive system architect and collaborative developer focused on writing efficient, scalable code to solve actual problems.

Education

JIS College of Engineering | *B.Tech in Computer Science and Engineering* 2025 – 2029
– **Academic Standing:** 1st Semester GPA: **8.7 CGPA** (Pursuing)

Skills & Achievements

- **Languages:** Java, C/C++, C, Python
- **Core Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Role-Based Access Control (RBAC)
- **Applied Experience & Tools:** Full-Stack Web Development (HTML5, CSS3, JavaScript), FastAPI, MongoDB, Git, GitHub
- **AI Integration:** OpenAI API, Google Grok API
- **Hardware/IoT:** Arduino Uno, ESP32, Raspberry Pi, Sensor Integration
- **Achievements:** Active Hackathon Competitor (Participant in Crazy Build Hackathon)

Top Projects

- AI-Enabled Business Analytics System** | *HTML, CSS, JS, Python, MongoDB, LLM* Solo Project
- **Data Parsing & Ingestion:** Engineered an automated backend engine to ingest and read enterprise reports directly from uploaded Excel (.xlsx) and CSV files.
 - **Automated Intelligence:** Programmed modules to analyze historical business growth, compute profit margins, and identify micro-trends or dips in specific sales sectors.
 - **LLM Recommendations:** Integrated generative AI pipelines to evaluate parsed sheet data and automatically generate contextual strategic expansion recommendations.
 - **Interactive Analytics Suite:** Built a comprehensive user dashboard featuring real-time health metrics, interactive growth charts, and an automated function to download full physical reports.
- Monolithic SQL-Backed Secure Banking System** | *C, SQL Database, Security Algorithms* Solo Project
- **Fraud Detection Mechanism:** Developed an automated algorithmic guardrails system that scans transactions and instantly locks an account if high-volume anomalies (1,00,000 or above) are triggered.
 - **Multi-Factor Access Routing:** Implemented strict credential verification workflows that dynamically generate secondary One-Time Passwords (OTP) to prevent unauthorized account access or credential-bypassing attempts.
- AI-Powered Celebratory & Greeting Portal** | *Full-Stack Frameworks, Generative AI APIs* Solo Project
- **Contextual Generation Engine:** Developed a web application that evaluates multi-user variables (names, relationships, specific occasions) to generate unique, highly personalized thematic letters or messages.
 - **Media Editing Canvas:** Designed a responsive interface allowing users to dynamically upload media assets, embed custom images, preview generated layouts, and download the finished stylized letter.

Interests

- Participating in fast-paced Hackathons and collaborative tech events
- Full-Stack Web Architecture & Cybersecurity Concepts
- Exploring Advanced Data Structures, File I/O Systems, and Systems Programming